UNIVERSIDADE ESTADUAL DE MONTES CLAROS – UNIMONTES
CENTRO DE CIÊNCIAS EXATAS E TECNOLÓGICAS – CCET
DEPARTAMENTO DE CIÊNCIAS DA COMPUTAÇÃO – DCC

ATIVIDADE AVALIATIVA II

RAMON LOPES DE QUEIROZ

MONTES CLAROS – MG
ABRIL/2026

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Atividade avaliativa apresentada para atendimento de requisito parcial para aprovação na disciplina Geometria Analítica e Álgebra Linear do Curso de Graduação em Bacharelado em Sistemas de Informação – 2º período

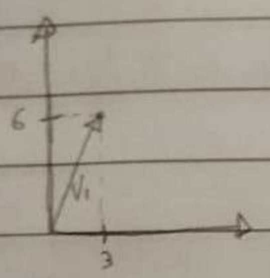
Professor: Antônio Wilson Vieira

MONTES CLAROS – MG

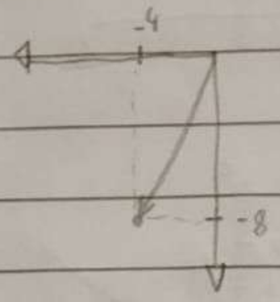
ABRIL/2026

Exercício - Secção 3.1 - 3, 5, 7, 9, 13, 26, 31 -

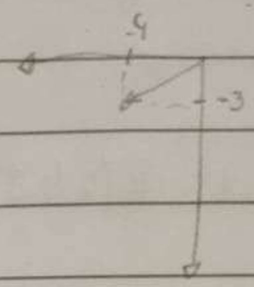
3- a- $V_1 = (3,6)$



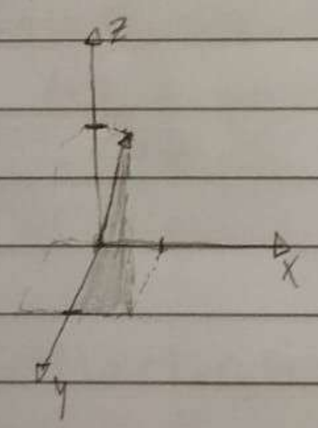
b- $V_2 = (-4,-8)$



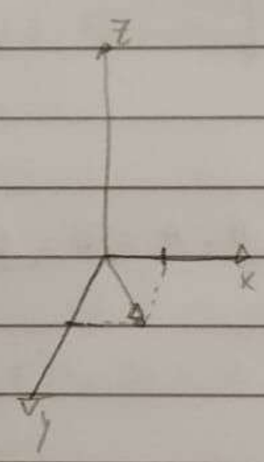
c- $V_3 = (-4,-3)$



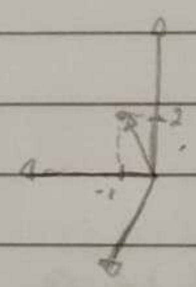
d- $V_4 = (3,4,5)$



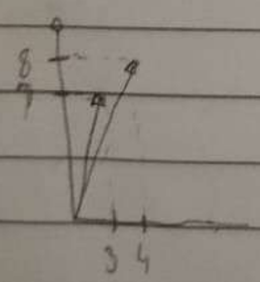
e- $V_5 = (3,3,0)$



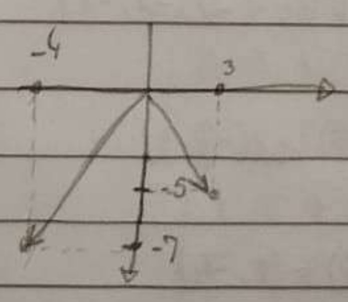
f- $V_6 = (-1,0,2)$



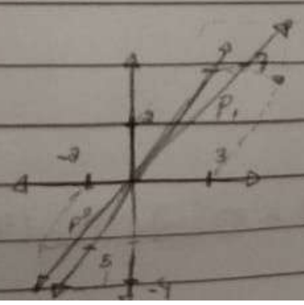
5- a- $P_1(4,8)$, $P_2(3,7)$



b- $P_1(3,-5)$, $P_2(-4,-7)$



c- $P_1(3,-7,2)$, $P_2(-2,5,-4)$



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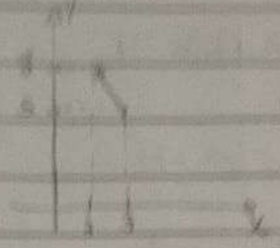
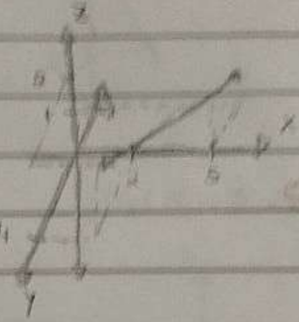
7- a. $P_1(3,5)$, $P_2(2,8)$



b- $P_1(5,-2,1)$, $P_2(2,4,2)$



9a. $A(1,1)$, $U(1,2)$ $u = B - A \Rightarrow B = u + A$

$V = (1+1, 1+2) \Rightarrow V = (2,3)$

b. $B(-1,-1,2)$, $U(1,1,3)$ $A = B - u$

$V = (-1-1, -1-1, 2-3) \Rightarrow V = (-2,-2,-1)$

13- a. $u = (4,-1)$, $v(0,5)$, $w(-3,-3)$

$u + w = (4,-1) + (-3,-3) = (1,-4)$

b. $(0,5) - 3.(4,-1)$
$(-12, 8)$

c. $2u - 10w = 2(4,-1) - 10(-3,-3) = (8,-2) - (-30,-30) = (38,22)$

d- $3v - 2(u + 2w) = (4,-1) + 2(-3,-3) + 3v$

$3(0,5) - 2(-2,-7)$

$(4,29)$

e- $-3(w - 2u + v) = 3((-3,-3) - 2(4,-1) + (0,5))$

$-3(-11,4)$

$(33,-12)$

f. $(-2u - v) - 5(v + 3w)$

$-2(4,-1) - (0,5) - 5((0,5) + 3(-3,-3))$

$(-8,-3) - (45,-20)$

$(37,17)$

26- $c_1(1,2,0) + c_2(2,1,1) + c_3(0,3,1) = (0,0,0)$

$$\begin{cases} a + 2b + 0 = 0 \\ 2a + b + 3c = 0 \\ 0 + b + c = 0 \end{cases} \Rightarrow \begin{matrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 3 & 0 \\ 0 & 1 & 1 & 0 \end{matrix}$$

$c_1 = 0$

$c_2 = 0$

$c_3 = 0$

São linearmente independentes.

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$b = -c$

$a + 2(-c) = 0 = -2c$

$2(-2c) - c + 3c = 0$

$c = 0$

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31- $c_1(-2,9,6)+c_2(-3,2,1)+c_3(1,7,5)=(0,5,4)$

$$\begin{cases} -2a-3b+c=0 \\ 9a+2b+7c=5 \\ 6a+b+5c=4 \end{cases} \Rightarrow \begin{bmatrix} -2 & -3 & 1 & 0 \\ 9 & 2 & 7 & 5 \\ 6 & 1 & 5 & 4 \end{bmatrix} \xrightarrow{-\frac{1}{2}L_1} \begin{bmatrix} 1 & 3/2 & -1/2 & 0 \\ 9 & 2 & 7 & 5 \\ 6 & 1 & 5 & 4 \end{bmatrix} \xrightarrow[L_3-6L_1]{L_2-9L_1} \begin{bmatrix} 1 & 3/2 & -1/2 & 0 \\ 0 & -23/2 & 23/2 & 5 \\ 0 & -8 & 8 & 4 \end{bmatrix} \xrightarrow{-\frac{2}{23}L_2}$$

$$\begin{bmatrix} 1 & 3/2 & -1/2 & 0 \\ 0 & 1 & -1 & -10/23 \\ 0 & -8 & 8 & 4 \end{bmatrix} \xrightarrow{L_3+8L_2} \begin{bmatrix} 1 & 3/2 & -1/2 & 0 \\ 0 & 1 & -1 & -10/23 \\ 0 & 0 & 0 & 12/23 \end{bmatrix}$$

SI → Não tem escalares possíveis.

Exercício - Secção 3.2 (Ímpares) -

1- a - $V = (4,-3)$ $\|V\| = \sqrt{4^2+(-3)^2} = \sqrt{25} = 5$

$u = \frac{1}{\|\vec{V}\|} \cdot v = \frac{1}{5} \cdot (4,-3) = \left(\frac{4}{5}, \frac{-3}{5}\right)$

$-u = -\left(\frac{4}{5}, \frac{-3}{5}\right) = \left(\frac{-4}{5}, \frac{3}{5}\right)$

b - $V = (2,2,2)$ $\|V\| = \sqrt{2^2+2^2+2^2} = \sqrt{12} \cong 3,46 = 2\sqrt{3}$

$u = \frac{1}{2\sqrt{3}} (2,2,2) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

$-u = \left(\frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$

c - $V = (1,0,2,1,3)$ $\|V\| = \sqrt{1^2+0^2+2^2+1^2+3^2} = \sqrt{15}$

$u = \left(\frac{1}{\sqrt{15}}, 0, \frac{2}{\sqrt{15}}, \frac{1}{\sqrt{15}}, \frac{3}{\sqrt{15}}\right)$

$-u = \left(\frac{-1}{\sqrt{15}}, 0, \frac{-2}{\sqrt{15}}, \frac{-1}{\sqrt{15}}, \frac{-3}{\sqrt{15}}\right)$

3. a - $u = (2,-2,3)$ $V = (1,-3,4)$ $W = (3,6,-4)$

$u+V = (2,-2,3) + (1,-3,4)$
$(3,-5,7)$

$\|u+V\| = \sqrt{3^2+(-5)^2+7^2} = \sqrt{83}$

b - $\|u\| + \|V\|$
$\sqrt{2^2+(-2)^2+3^2} + \sqrt{1^2+(-3)^2+4^2}$
$\sqrt{17} + \sqrt{26}$

c - $\|-2u+2V\| = 2\|-u+V\|$
$2 \cdot \sqrt{(-1)^2+(-1)^2+(1)^2}$
$2\sqrt{3}$

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d $\|3u - 5v + w\|$ $\quad \|(4,15,-15)\| = \sqrt{4^2+15^2+(-15)^2} = \sqrt{466}$

$3.(2,-2,3) - 5(1,-3,4) + (3,6,-4)$

$(6,-6,9) - (5,-15,20) + (3,6,-4)$

$(4,15,-15)$

5 - a. $u = (-2,-1,4,5)$ $v = (3,1,-5,7)$ $w = (-6,2,1,1)$

$3.(-2,-1,4,5) - 5.(3,1,-5,7) + (-6,2,1,1)$

$(-6,-3,12,15) - (15,5,-25,35) + (-6,2,1,1)$

$(-27,-6,38,-19)$

$\|(-27,-6,38,-19)\| = \sqrt{(-27)^2+(-6)^2+(38)^2+(-19)^2} = \sqrt{729+36+1444+361} = \sqrt{2570}$

b. $\|3u\| - 5\|v\| + \|w\|$

$\|-6,-3,4,5\| - 5\|3,1,-5,7\| + \|-6,2,1,1\|$

$\sqrt{86} - 5\sqrt{84} + \sqrt{42}$

c. $\| - \|u\|.v\|$

$\| - \|2,-1,4,5\|.(3,1,-5,7)\|$

$\| -(\sqrt{46}).(3,1,-5,7)\|$

$\| -3\sqrt{46}, -\sqrt{46}, +5\sqrt{46}, -7\sqrt{46}\|$

$\sqrt{(-3\sqrt{46})^2 + (-\sqrt{46})^2 + (5\sqrt{46})^2 + (-7\sqrt{46})^2}$

$\sqrt{414 + 46 + 1150 + 2254}$

$\sqrt{3864}$

7. $V = (-2, 3, 0, 6)$ $\quad \|V\| = \sqrt{(-2)^2 + 3^2 + 0^2 + 6^2} = \sqrt{49} = 7$

$\|KV\| = 5$

$K \cdot \|V\| = 5$

$K \cdot 7 = 5$

$K = \pm \frac{5}{7}$

9. a. $(3,1,4) \cdot (2,2,-4)$ $\quad (3,1,4) \cdot (3,1,4)$ $\quad (2,2,-4) \cdot (2,2,-4)$

$(3 \cdot 2, 1 \cdot 2, 4 \cdot -4)$ $\quad (9, 1, 16)$ $\quad (4, 4, 16)$

$(6, 2, -16)$ $\quad 9 + 1 + 16$ $\quad 24$

$6 + 2 - 16$ $\quad 26$

$P_E = -8$

soma de produtos

b. $(1,1,4,6) \cdot (2,-2,3,-2)$ $\quad (1,1,4,6) \cdot (1,1,4,6)$ $\quad (2,-2,3,-2) \cdot (2,-2,3,-2)$

$2 - 2 + 12 - 12$ $\quad 1, 1, 16, 36$ $\quad 4 + 4 + 9 + 4$

0 $\quad 54$ $\quad 21$

11 a. $u = (3,3,3)$, $V = (1,0,4)$

$\sqrt{(3-1)^2 + (3-0)^2 + (3-4)^2}$

$\sqrt{2^2 + 3^2 + (-1)^2}$

$\sqrt{14}$

b. $u = (0,-2,-1,1)$, $v = (-3,2,4,4)$

$\sqrt{(0+3)^2 + (-2-2)^2 + (-1-4)^2 + (1-4)^2}$

$\sqrt{9 + 16 + 25 + 9}$

$\sqrt{59}$

c. $u = (3,-3,-2,0,3,13,5)$ $\quad v = (-4,1,-1,5,0,-11,4)$

$\sqrt{(3+4)^2 + (-3-1)^2 + (-2+1)^2 + (0-5)^2 + (3-0)^2 + (13+11)^2 + (5-4)^2}$

$49 + 16 + 1 + 25 + 9 + 576 + 1$

$\sqrt{677}$

animativa

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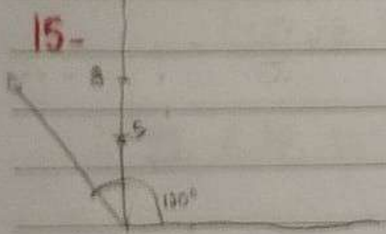
13. a. $\cos\theta = \frac{u \cdot v}{\|u\| \cdot \|v\|} = \frac{(3.1)+(3.0)+(3.4)}{(\sqrt{9+9+9}) \cdot \sqrt{1+0+16}} = \frac{15}{\sqrt{27} \cdot \sqrt{17}} = \frac{15}{3\sqrt{3} \cdot \sqrt{17}}$ = Positivo = ângulo agudo

b. $\cos\theta = \frac{(0,-3)+(-2,2)+(-1,4)+(1,4)}{\sqrt{0+4+1+1} \cdot \sqrt{9+4+16+16}} = \frac{-4}{\sqrt{6} \cdot \sqrt{45}}$ = Negativo = Obtuso

c. $\cos\theta = \frac{(3,-4)+(-2,1)+(-2,-1)+(0,5)+(-3,0)+(13,-11)+(5,4)}{\sqrt{9+9+4+0+9+169+25} \cdot \sqrt{16+1+1+25+0+121+16}} = \frac{-136}{\sqrt{225} \cdot \sqrt{180}}$ = Negativo = Obtuso

15.



$|a| = 9$, $120°$ $120 - 90 = 30°$

$|b| = 5$, $90°$

$9 . 5 . \cos(30°)$

$45 . \frac{\sqrt{3}}{2}$

$a . b = 22,5\sqrt{3} \cong 38,97$

17. a. $u.(v.w)$ = Errado. $v.w$ resulta em uma escalar, e a multiplicação escalar só acontece entre dois vetores, e não entre escalar e vetor.

b. $u.(v+w)$ = Certo.

c. $\|u.v\|$ = Não. Pois a norma é pra vetores, e a multiplicação resulta em uma escalar.

d. $(u.u) - \|w\|$ = Certo.

19- a. $(-4,-3) = \sqrt{(-4)^2+(-3)^2} = \sqrt{25} = 5$ $\quad u = \left(\frac{-4}{5}, \frac{-3}{5}\right)$

b. $(1,7) = \sqrt{1+49} = \sqrt{50}$ $\quad u = \left(\frac{1}{\sqrt{50}}, \frac{7}{\sqrt{50}}\right)$

c. $(-3,2,\sqrt{3}) = \sqrt{(-3)^2+2^2+\sqrt{3}^2} = \sqrt{16} = 4$ $\quad u = \left(-\frac{3}{4}, \frac{2}{4}, \frac{\sqrt{3}}{4}\right)$

d. $(1,2,3,4,5) = \sqrt{1^2+2^2+3^2+4^2+5^2} = \sqrt{55}$ $\quad u = \left(\frac{1}{\sqrt{55}}, \frac{2}{\sqrt{55}}, \frac{3}{\sqrt{55}}, \frac{4}{\sqrt{55}}, \frac{5}{\sqrt{55}}\right)$

21- $u = \frac{v}{\|v\|}$ $\quad w = m \cdot u$

$w = m \cdot \frac{v}{\|v\|}$

23- a. $u = (2,3)$, $v = (5,-7)$

$\frac{2.5 + 3.-7}{\sqrt{2^2+3^2} \cdot \sqrt{5^2+(-7)^2}} = \frac{-11}{\sqrt{13} \cdot \sqrt{74}} = \frac{-11}{\sqrt{962}}$

b. $\cos\theta = \frac{-6.4+(-2.0)}{\sqrt{36+4} \cdot \sqrt{16+0}} = \frac{-24}{\sqrt{40} \cdot \sqrt{16}} = \frac{-24}{4\sqrt{40}}$

c. $\cos\theta = \frac{1.3+5.-3+4.3}{\|u\| \cdot \|v\|} = \frac{0}{\|u\| \|v\|} \rightarrow \cos\theta = 0$

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d. $\cos\theta = -2.1 + 2.7 + 3.-4 = 0 \rightarrow \cos\theta = 0$

25\. a. $u = (3,2)$ $v = (4,-1)$

$|3.4 + 2.-1| \leq \sqrt{3^2+2^2} \cdot \sqrt{4^2+(-1)^2}$

$|10| \leq \sqrt{221}$ ✓

b. $u(-3,1,0)$, $v(2,-1,3)$

$|-3.2 + 1.-1 + 0.3| \leq \sqrt{-3^2+1^2+0^2} \cdot \sqrt{2^2+(-1)^2+3^2}$

$7 \leq \sqrt{10} \cdot \sqrt{14}$

$7 \leq \sqrt{140}$ ✓

c. $u = (0,2,2,1)$, $v = (1,1,1,1)$

$|0.1 + 2.1 + 2.1 + 1.1| \leq \sqrt{0^2+2^2+2^2+1^2} \cdot \sqrt{1^2+1^2+1^2+1^2}$

$5 \leq \sqrt{9} \cdot \sqrt{4}$

$5 \leq 6$ ✓

27\. $\|P - P_0\| = 1$ $P - P_0 = (x - x_0, y - y_0, z - z_0)$

$(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = 1^2$

Centro = $P_0 = (x_0, y_0, z_0)$.

Raio = $r = 1$.

$S = \{(x,y,z) \in \mathbb{R}^3 \mid (x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = 1\}$

29\. $\|v\| = \sqrt{v_1^2 + v_2^2 + \ldots + v_n^2}$

a.

$\|V\| \geq 0$, Como o quadrado sempre vai ser +, a raiz e a soma também, então $\|V\| \geq 0$.

b. $\|V\| = 0$, se, e só se, $v = 0$.

Se $v \neq 0$, a soma sempre vai ser maior que 0. O único caso é $V = 0$.

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31.d. $(u - v) \cdot w = u \cdot w - v \cdot w$ $\qquad$ $(u-v) \cdot w = \sum_{i=1}^{n} (u_i - v_i) w_i = \sum_{i=1}^{n} (u_i w_i - v_i w_i)$

$(uw - vw)$ → Distributividade $\qquad$ $\sum u_i w_i - \sum v_i w_i$

e - $a(u \cdot v) = u(a v)$

$a(u_1 v_1 + u_2 v_2 + \dots + u_n v_n)$

$a u_1 v_1 + a u_2 v_2 + \dots + a u_n v_n$

$u_1(a v_1) + u_2(a v_2) + \dots + u(a v_n)$

33- $\|u + v\| \leq \|u\| + \|v\|$

$(\|u + v\|)^2 = (\|u\| + \|v\|)^2$

$\cancel{\|u\|^2} + 2(u \cdot v) + \cancel{\|v\|^2} = \cancel{\|u\|^2} + 2\|u\|\|v\| + \cancel{\|v\|^2}$

$2(u \cdot v) = 2\|u\|\|v\|$

$u \cdot v = \|u\|\|v\|$

$\|u\|\|v\| \cos\theta = \|u\|\|v\|$

$\cos\theta = 1$ $\qquad$ $\theta = 0°$ $\qquad$ Vetores paralelos com o mesmo sentido.

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Exercício 3.3 - (1, 4, 7, 10, 13, 19, 29, 33, 41) -

1a - $u = (6, 1, 4)$ $\quad (6,1,4)\cdot(2,0,-3)$ $\quad$ Ortogonal.
$V = (2, 0, -3)$

$12 + 0 - 12$

0

b - $u = (0, 0, -1)$ $\quad 0\cdot 1 + 0\cdot 1 + (-1\cdot 1)$ $\quad$ Não é ortogonal.
$V = (1, 1, 1)$ $\quad -1$

c - $u = (-6, 0, 4)$ $\quad (-6\cdot 3) + (0\cdot 1) + (4\cdot 6)$ $\quad$ Não é ortogonal.
$V = (3, 1, 6)$ $\quad -18 + 24$

6

d - $u = (2, 4, -8)$ $\quad (2\cdot 5) + (4\cdot 3) + (-8\cdot 7)$ $\quad$ Não é ortogonal.
$v = (5, 3, 7)$ $\quad -56 + 22$

-34

4 - a - $V_1 = (2, 3)$ $\quad (2\cdot -3) + (3\cdot 2)$ $\quad$ Conjunto ortogonal.
$V_2 = (-3, 2)$ $\quad 0$

b - $V_1 = (1, -2)$ $\quad (1\cdot -2) + (-2\cdot 1)$ $\quad$ Não é um conjunto ortogonal.
$V_2 = (-2, 1)$ $\quad -2 - 2 = -4$

c - $V_1 = (1, 0, 1)$ $\quad (1\cdot 1\cdot -1) + (0\cdot 1\cdot 0) + (1\cdot 1\cdot 1)$ $\quad$ É um conjunto ortogonal.
0
$V_2 = (1, 1, 1)$
$V_3 = (-1, 0, 1)$

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d. $V_1(2,-2,1)$ $(2,2,1)+(-2,1,2)+(1,-2,2)$ Não é um conjunto ortogonal.

$V_2(2,1,-2)$ -4

$V_3(1,2,2)$

7. $A=(1,1,1)$ $d^2=\Delta x^2+\Delta y^2+\Delta z^2$

$B=(-2,0,3)$

$C=(-3,-1,1)$

$AB = x = -2-1=-3$ $(-3)^2+(-1)^2+(2)^2=14$

$y=0-1=-1$

$z=3-1=2$

catetos

$BC = x=-3+2=-1$ $1^2+(-1)^2+(-2)^2=6$

$y=-1-0=-1$

$z=1-3=-2$

Hipotenusa

$AC = x=-3-1=-4$ $(-4)^2+(-2)^2+0^2=20$

$y=-1-1=-2$

$z=1-1=0$

Sim. Como a soma do quadrado dos catetos (14 e 6) é igual o quadrado da hipotenusa (20).

$a^2+b^2=c^2$

$14+6=20$

10. $P(1,1,4)$; $n(1,9,8)$

→ Forma Ponto-normal

$1(x-1)+9(y-1)+8(z-4)=0$

$x-1+9y-9+8z-32=0$

$x+9y+8z-42=0$

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13- $4x - y + 2z = 5$ $\quad 7x - 3y + 4z = 8$

$n = (4, -1, 2)$ $\quad n = (7, -3, 4)$

Razão $x = \frac{4}{7}$ $\quad$ Razão $y = \frac{1}{3}$ $\quad$ Razão $z = \frac{1}{2}$ $\quad \frac{4}{7} \neq \frac{1}{3} \neq \frac{1}{2}$ / Não são paralelas.

19- $\|proj_{(-4,-3)}(1,-2)\| = \frac{|u \cdot a|}{\|a\|}$ $\quad \frac{|(1 \cdot -4) + (-2 \cdot -3)|}{\sqrt{(-4)^2 + (-3)^2}} = \frac{+2}{5} = 0{,}4.$

29- $4x + 3y + 4 = 0$
$(-3, 1)$ $\quad \frac{|\overset{-12}{4 \cdot (-3)} + \overset{+3}{(+3 \cdot 1)} + \overset{+4}{4}|}{\sqrt{4^2 + (-3)^2}} = \frac{5}{5} = 1$ $\quad d = 1.$

33- $x + 2y - 2z = 4$
$(3, 1, -2)$ $\quad \frac{|3 + 2 \cdot (1) - 2 \cdot (-2) - 4|}{\sqrt{1^2 + 2^2 + (-2)^2}} = \frac{5}{3}$ $\quad d = \frac{5}{3}$

41- a. $\cos\alpha = \frac{a}{\|v\|}$ $\quad V = (a, b, c)$ $\quad i = (1,0,0) = a$

$V \cdot i = \|V\| \|i\| \cos\alpha$

$\|i\| = 1$

$a = \|V\| (1) . \cos\alpha$

$\cos\theta = \frac{a}{\|v\|}.$

b. $\beta = V \cdot j = b$ $\quad b = \|V\| . 1 . \cos\beta \Rightarrow \cos\beta = \frac{b}{\|v\|}.$

$\gamma = V \cdot K = c$ $\quad c = \|V\| . (1) . \cos\gamma \rightarrow \cos\gamma = \frac{c}{\|v\|}.$

c- $V = (a, b, c)$

$$\frac{V}{\|V\|} = \left(\frac{a}{\|V\|}, \frac{b}{\|V\|}, \frac{c}{\|V\|}\right)$$

$$\frac{V}{\|V\|} = (\cos\alpha, \cos\beta, \cos\gamma).$$

d- $(x, y, z) = x^2 + y^2 + z^2$

$\|u\|^2 = \cos^2\alpha + \cos^2\beta + \cos^2\gamma$

$\|u\|^2 = 1$

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$